

OIL INDIA LIMITED - eRTMAC OPERATIONS

OFFICIAL WELL COMPLETION REPORT (WCR) & DRILLING HISTORY
WELL IDENTIFIER: 15/9-14

1. Well Location & Technical Header

Operator: Oil India Limited / Joint North Sea	Well Name: 15/9-14
UTM Easting (X): 423247.1 m	UTM Northing (Y): 6461858.2 m
Latitude: 58.291030° N	Longitude: 1.690818° E
Total Depth (MD): 3564.6 m	Spud Depth: 480.6 m
Drilling Rig: eRTMAC Semi-Submersible Rig-07	Status: Plugged & Abandoned (P&A;) / Offset Reference

2. Stratigraphic Progression & Formation Tops

Formation / Stratum Name	Top Depth MD (m)	Lithology Characterization	Drillability / Risk Profile
Utsira Fm.	480.6 m	Interbedded Sand / Shale	Loss / Depletion
Balder Fm.	634.8 m	Interbedded Sand / Shale	Loss / Depletion
Sele Fm.	789.0 m	Shale / Claystone with interbedded siltstone	Tight Hole / Swelling
Lista Fm.	943.2 m	Interbedded Sand / Shale	Loss / Depletion
Heimdal Fm.	1097.4 m	Porous Turbidite Sandstone (Hydrocarbon Bearing)	Loss / Depletion
Ty Fm.	1251.6 m	Porous Turbidite Sandstone (Hydrocarbon Bearing)	Loss / Depletion
Ekofisk Fm.	1405.8 m	Chalk / Hard Calcareous Limestone	Fractured / Losses
Tor Fm.	1560.0 m	Chalk / Hard Calcareous Limestone	Fractured / Losses
Hod Fm.	1714.2 m	Interbedded Sand / Shale	Loss / Depletion
Blodoeks Fm.	1868.4 m	Interbedded Sand / Shale	Loss / Depletion
Svarte Fm.	2022.6 m	Interbedded Sand / Shale	Loss / Depletion
Roedby Fm.	2176.8 m	Interbedded Sand / Shale	Loss / Depletion
Sola Fm.	2331.0 m	Interbedded Sand / Shale	Loss / Depletion
Asgard Fm.	2485.2 m	Interbedded Sand / Shale	Loss / Depletion
Draupne Fm.	2639.4 m	Interbedded Sand / Shale	Loss / Depletion
Heather Fm.	2793.6 m	Interbedded Sand / Shale	Loss / Depletion
Hugin Fm.	2947.8 m	Interbedded Sand / Shale	Loss / Depletion
Sleipner Fm.	3102.0 m	Interbedded Sand / Shale	Loss / Depletion
Skagerrak Fm.	3256.2 m	Interbedded Sand / Shale	Loss / Depletion
Smith Bank Fm.	3410.4 m	Interbedded Sand / Shale	Loss / Depletion

3. Casing Design & Mud Program Summary

36 in Hole	30 in Conductor	350 m	Spud Mud / Seawater	1.05 SG	FIT 1.25 SG
26 in Hole	20 in Surface	1,050 m	KCl / Polymer	1.12 SG	LOT 1.45 SG
17-1/2 in Hole	13-3/8 in Interm.	2,200 m	Low Solids Non-Dispersed	1.24 SG	LOT 1.62 SG
12-1/4 in Hole	9-5/8 in Prod.	3100 m	Oil Based / Synthetic Mud	1.32 SG	LOT 1.78 SG
8-1/2 in Hole	7 in Liner	3565 m	Synthetic Oil Based (SOBM)	1.38 SG	LOT 1.88 SG

4. Historical Operational Incidents, NPT & Mitigations (Well: 15/9-14)

This section constitutes the institutional memory log extracted by the NWIS knowledge engine. All alerts generated during active drilling cite these exact records.

INCIDENT #1: Well Kick / Influx	Severity: CRITICAL	NPT Hours: 32.0 hrs
Depth Interval: 2150.0 m - 2185.0 m	Formation: ROGALAND GP	Status: Resolved on Site
Event Occurrence Narrative: Drilling break noted at 2162m with ROP increasing from 12 m/hr to 38 m/hr. Pit gain of 18 bbls observed within 4 minutes. Active gas detector spiked from 1.2% to 14.8% methane.		
Mitigation Action Applied: Immediately performed space-out, shut down mud pumps, and closed annular blowout preventer (BOP). SIDPP stabilized at 320 psi; SICP stabilized at 410 psi. Executed Wait and Weight kill sheet procedure, increasing mud weight from 1.22 SG to 1.34 SG. Circulated gas bubble safely through choke manifold and degasser.		
Offset Drilling Recommendation & Institutional Lesson: Rogaland gas-charged sand stringers exhibit sharp abnormal pressure ramps. Implement flow check at any ROP break exceeding 25 m/hr.		

INCIDENT #2: Borehole Instability / Tight Hole	Severity: MEDIUM	NPT Hours: 12.0 hrs
Depth Interval: 2310.0 m - 2360.0 m	Formation: SELE FM	Status: Resolved on Site
Event Occurrence Narrative: Severe tight hole encountered on trip out at 2335m. Caliper log revealed extensive hole enlargement (16 inch in 12-1/4 inch section) and cavings at shaker.		
Mitigation Action Applied: Performed back-reaming with high pump rate. Pumped 30 bbl tandem high-density sweeps (1.30 SG) to evacuate reactive shale cavings. Raised potassium chloride (KCl) concentration to 7% wt to inhibit clay swelling.		
Offset Drilling Recommendation & Institutional Lesson: Sele Formation reactive smectite shales require stringent filtrate loss control (< 4.0 cc) and minimum 6% KCl inhibition.		